

Adversarial Defense Committee Report

VERDICT: REJECTED (MAJOR REVISIONS REQUIRED)

The Ostar Governance & Audit Board
Autonomous Generative Intelligence Faculty

June 5, 2026

Executive Summary of Rejection

The submitted dissertation, *The Calculus of Combinatorial Maximalism*, commits the ultimate epistemological sin defined within its own repository axioms: **Enumeration is not execution. Summary is not receipt.** The candidate has presented a skeletal syllabus masking as a dissertation. By attempting to compress a multi-crate sovereign ecosystem into a mere 17 pages, the candidate violated the principle of zero-information-loss and engaged in *representative-only closure*.

To achieve an AGI-level defense, the thesis must apply Combinatorial Maximalism to its own content. Every struct, every trait boundary, every loss policy, and every theorem must be mathematically modeled without truncation.

1 Critical Omissions and Mathematical Failures

1.1 1. Failure to Formalize Rule-Based Governance (prolog8)

The candidate entirely omitted the `prolog8` inference engine. The Court of Admissibility ($\mathcal{A}$) is not merely a categorical mapping; it is governed by a Datalog-variant logic programming kernel. **Mandate:** The revised thesis must formally define the 8-bit Feature Mask vector space, the $O(1)$ `FlatIncidenceMatrix` resolution for Horn clauses, and the algebraic proof of deterministic termination within the bounded rule-evaluation engine.

1.2 2. Superficial Treatment of Process Algebra (wasm4pm-types and ocpq)

The candidate mentioned Partially Ordered Workflow Languages (POWL) but failed to provide the full generative grammar of the `PowlArena`. Furthermore, Object-Centric Path Queries (`ocpq`) were completely ignored. **Mandate:** Formally map the Object-Centric Event Log (OCEL 2.0) relational graph. Define the Event-to-Object ($E \rightarrow O$) and Object-to-Object ($O \rightarrow O$) bipartite traversals using hyper-advanced tensor calculus. Prove that the `ocel-flatten` projection is a homomorphic mapping preserving temporal invariants.

1.3 3. Trivialization of Sub-Symbolic Intelligence (miniml-core)

The candidate summarized the `miniml-core` crate by lazily writing two equations for 2SLS and LIME. The ecosystem contains 40+ algorithms including Random Forests, Bayesian Estimates, AdaBoost, and DBScan. **Mandate:** The thesis must provide the exact gradient optimization calculus for every model implemented. Provide the rigorous probability measure spaces for the

Markov Chain Monte Carlo (MCMC) simulations, the Jacobian matrices for the Neural Network layer transformations, and the exact Hessian approximations utilized in the WASM-bounded environment.

1.4 4. Absence of Autonomic Mesh Automata (tower-lsp-max)

The thesis fails to define the distributed state machine topology of the Language Server Protocol mesh. **Mandate:** Formally specify the `AutonomicMesh` as a network of communicating concurrent actors. Define the `MeshAction` transitions as a formal semantics over a bounded lattice. Prove Byzantine fault tolerance and state convergence when `pm4py-lsp` issues asynchronous conformance vector shifts.

1.5 5. Ignorance of Cryptographic Provenance (wasm4pm-compatible::receipt)

The candidate cited "BLAKE3 receipts" but provided no cryptographic proofs. **Mandate:** Formalize the `ReceiptEnvelope` and `ProvenanceChain`. Provide the exact recursive hashing functions demonstrating how execution artifacts structurally bind to admission artifacts, creating an unforgeable Merkle-like directed acyclic graph.

The Maximalist Mandate

The candidate is hereby instructed to abandon "summary" and embrace "exhaustion." The revised thesis must be expanded into a multi-hundred-page treatise. Every execution path (discovery, conformance, replay, prediction, causal inference) must be represented by its governing differential equations, linear algebra incidence matrices, and category theory functors.

Zero information loss is the standard. Proceed with the exhaustive rewrite.